A Simplified Subband ANC Algorithm Without Secondary Path Modeling

Min Gao, Jing Lu, and Xiaojun Qiu

***Abstract*—Active noise control algorithms without secondary path modeling are very appealing because they can effectively overcome the problems caused by imperfect secondary path model. For broadband noise control, subband structures have been adopted in these kinds of algorithms, where the full-band adaptive filter coefficients are obtained from subband filter coefficients with weight transformation. However, the usually used two-direction search method suffers from slow convergence if the secondary path phase is close to $\pm 90^\circ$. To deal with this problem, a four-direction search method has been proposed in frequency domain, which is sometimes too complicated for practical applications. In this paper, a simplified algorithm is proposed to reduce the implementation complexity. In the proposed algorithm, the delay of the secondary path is estimated first, and then the estimated delay is used in the delay compensation to reduce the number of subbands. Two reference signals are generated in each subband and two update directions are used. Only one subband reference signal and one update direction are selected to approximate the phase response of the residual secondary path, and then the full-band adaptive controller coefficients are adjusted directly in time domain with them without the need of further weight transformation. Just like those existing subband algorithms, the simplified algorithm can obtain similar steady-state noise reduction performance as that of the FxLMS algorithm but without secondary path modeling. Simulations and experiments are carried out to demonstrate the feasibility of the proposed algorithm.**

***Index Terms*—Active noise control (ANC), adaptive controller, secondary path modeling, simplified subband algorithm.**

I. Introduction

ACTIVE noise control (ANC) is a method for attenuating unwanted disturbances by introduction of controllable secondary sources, whose outputs are arranged to interfere destructively with the original primary sound [1]–[3]. For most ANC algorithms, it is necessary to identify the secondary path transfer functions between the output of the adaptive controller and the error sensors [4]. The secondary path modeling can be roughly divided into off-line and on-line methods. Off-line modeling methods assume that the characteristics of secondary path are time-invariant and the secondary path modeling are usually conducted before the noise control process. On-line modeling methods [5]–[7] estimate the secondary path simultaneously but they often require an auxiliary noise input that contributes to the residual noise power. A delayed-x LMS algorithm was investigated by Kim *et al.*, where the secondary path model was approximated by a delay [8].

To avoid the secondary path identification, several ANC algorithms without secondary path identification have been proposed. The simultaneous equation method was proposed in Ref. 9 and the simultaneous perturbation method was proposed in Ref. 10. In Ref. 11, an updating path selection method was proposed, where the update direction of the adaptive controller has two choices, i.e., 0° and 180°, and the correct one is chosen automatically by monitoring the excess noise power. However, if the phase response of the secondary path is close to $\pm 90^\circ$, the algorithm converges very slowly, so a new frequency-domain delayless subband architecture has been proposed by the research group of the authors to solve the problem, where 4 potential update directions, i.e., 0°, 180°, and $\pm 90^\circ$, are used to find the best update direction [12]. The drawback of this method is that the filter updating process has to be carried out in frequency domain, leading to a high computational complexity. Therefore it is of significance to investigate a simplified method.

Most broadband noise control problems can be converted into narrowband ANC problems. Morgan *et al.* proposed a delayless subband adaptive filter architecture [13], where the input signal and error signal are divided into subband signals by using the polyphase FFT method. Park *et al.* proposed a frequency-domain implementation of the delayless subband filter algorithm to reduce the computational complexity of the delayless subband filter by decomposing the secondary path response into a set of subband functions [14]. Milani *et al.* proposed a new UDFTM-based adaptive subband filtering method to alleviate the degrading effects of the delay and side-lobe distortion introduced by the prototype filter on the system performance [15].

To avoid the time delay introduced by the subband-splitting, Schüldt *et al.* proposed a delayless subband adaptive filter partial updating scheme, where the idea is to only update the most misadjusted subband filter(s) [16]. The adaptive control filter is also divided into many subband filters, and then the full-band adaptive control filter coefficients are obtained with the weight transform methods [17]. The DFT-1 and DFT-2 stacking methods for weight transforming have nulls at the normalized frequency π and odd multiples of the normalized frequency π/M (M is the number of subbands), and the

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stacking distortion increases with the number of subbands [18]. The linear weight transform proposed by Larson *et al.* depends on the number of subbands, the decimation factor, the analysis filters and the synthesis filters, so the structure is complex with large computation complexity [19].

To avoid weight transformation, Courville *et al.* proposed a new subband structure, where the adaptive filter is not divided into subband filters and the full-band adaptive coefficients are updated directly by the subband signals [20]. This subband structure has been introduced in an ANC systems by DeBrunner *et al.*, where a filtered-x LMS algorithm based subband adaptive algorithm has been introduced with error path delay compensation to avoid the signal path delay [21]. In the frequency-domain subband architecture for ANC without secondary path modeling proposed by the group of the authors in 2009, the broadband noise control problem is converted into narrowband noise control problems [12]. The problem of the existing algorithms is that the full-band adaptive control filter coefficients need to be obtained with the weight transform methods. To reduce the complexity of the ANC system, Albu *et al.* proposed the multichannel pseudo-affine projection algorithm based on the dichotomous coordinate descent iterations [22], and the similar idea can be utilized in algorithm design in subband domain.

To simplify the existing algorithms for practical applications, a simplified subband active noise control algorithm (SSANC) without secondary path modeling is proposed in this paper, where the new features are:

1) time-domain implementation for full-band control filter updating and filtering;
2) the cost function is defined as the sum of the weighted squared subband errors for more control flexibility;
3) two narrowband reference signals, i.e., the delay compensated reference signal or orthogonal delay compensated reference signal, and two update directions, i.e., 0° or 180°, can be selected in each subband for updating the full-band control filter;
4) the full band control filter is not divided into several subband control filter, and updated directly by narrow band signals in time domain, which removes the necessity of using the weight transform methods;
5) the secondary path delay is roughly compensated in the algorithm to reduce the number of subbands in the implementation.

II. Simplified Subband ANC Algorithm Without Cancelation Path Modeling

Fig. 1 shows the diagram of the FxLMS algorithm, where $x(n)$ is the reference signal, $P(z)$ is the primary path transfer function between the primary noise $d(n)$ and $x(n)$, $W(z)$ is the adaptive controller; $s(n)$ is the actual control signal at the position of the error sensor; $\hat{C}(z)$ is a model of the true secondary path $C(z)$; $x_{\mathrm{f}}(n)$ is the filtered reference signal, and $v(n)$ is the additive noise.

The output of the adaptive controller $y(n)$ is obtained by filtering the reference signal $x(n)$ with the adaptive controller $W(z)$,

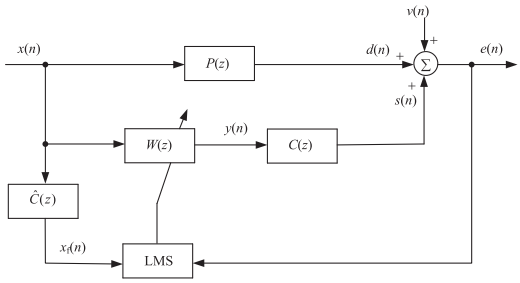


Fig. 1. The filtered-x LMS (FxLMS) algorithm.

$$y(n) = \sum_{l=0}^{N-1} w_l x(n-l) \tag{1}$$

where N is the length of the adaptive controller, $\{w_l\}$ are the coefficients of the adaptive controller $W(z)$. The actual control signal $s(n)$ can be expressed as,

$$s(n) = \sum_{i=0}^{N_{\mathrm{C}}-1} c_i y(n-i) \tag{2}$$

where N_{C} is the length of the secondary path, and $\{c_i\}$ are the coefficients of the impulse of the secondary path $C(z)$. The error signal $e(n)$ can be represented by,

$$e(n) = d(n) + s(n) + v(n) \tag{3}$$

where $v(n)$ represents the additive background noise.

Substituting Eq. (2) into Eq. (3), $e(n)$ can be written as,

$$e(n) = d(n) + v(n) + \sum_{l=0}^{N-1} w_l x_{\mathrm{f}}(n-l) \tag{4}$$

where the reference signal $x_{\mathrm{f}}(n)$ is the linear convolution of $x(n)$ with c_i. With perfect secondary path assumption $\hat{C}(z) = C(z)$, the filtered reference signal can be expressed as,

$$x_{\mathrm{f}}(n) = \sum_{i=0}^{N_{\mathrm{C}}-1} c_i x(n-i) \tag{5}$$

Based on the well-known LMS algorithm method, the adaptive coefficients are updated by [1],

$$w_l(n+1) = w_l(n) - \mu_{\mathrm{f}} e(n) x_{\mathrm{f}}(n-l) \tag{6}$$

where μ_{f} is the step size, and $l = 0, 1, \ldots, N-1$.

Fig. 2 shows the structure of the proposed simplified subband active noise control (SSANC) algorithm without secondary path modeling, which consists of 5 parts: (1) delay compensated reference signal generation, (2) orthogonal delay compensated reference signal generation, (3) subband signal generation, (4) adaptive weight update, and (5) subband filtered reference signal and update direction selection.

(1) Delay Compensated Reference Signal Generation: The signal $r(n)$ is obtained by filtering the reference signal $x(n)$, with a delay filter $z^{-\Delta}$, where Δ is the estimated sample number of the main delay of the secondary path. The estimation method of the direct path delay is detailed in the simulations and experiments section.

$$r(n) = x(n-\Delta) \tag{7}$$

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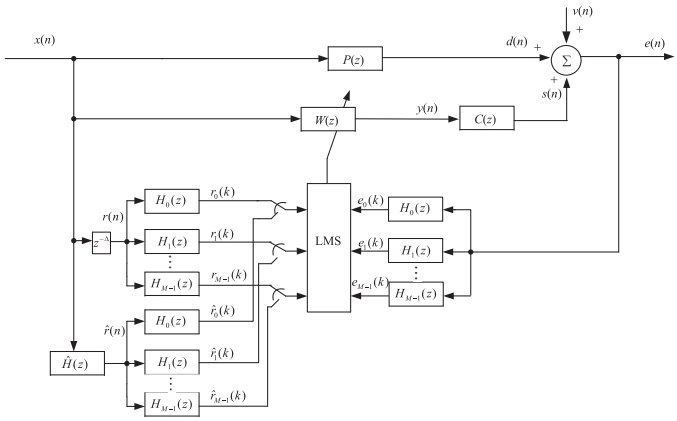


Fig. 2. The simplified subband ANC (SSANC) algorithm without secondary path modeling.

(2) Orthogonal Delay Compensated Reference Signal Generation: The orthogonal delay compensated reference signal, $\hat{r}(n)$, is obtained by filtering the reference signal $x(n)$ with an FIR filter $\hat{H}(z)$,

$$\hat{r}(n) = \sum_{i=0}^{L-1} h_i x(n-i) \tag{8}$$

where $\{h_i\}$ are the coefficients of the Hilbert filter $\hat{H}(z)$, and L is the length of Hilbert filter. The Hilbert filter has unity magnitude frequency response, a phase angle of $-\pi/2$ for $0 < \omega < \pi$, and a phase angle of $\pi/2$ for $-\pi < \omega < 0$ [21]. It is an ideal 90-degree phase shifter and can be designed simply with the firls() function in the MATLAB. In the pass-band of the Hilbert filter, the phase difference between $\hat{r}(n)$ and $r(n)$ is 90°, and the delay of Hilbert filter is designed to be the same as the delay Δ in equation (7) by letting $L = 2\Delta$.

(3) Subband Signal Generation: The delay compensated reference signal $r(n)$, the orthogonal delay compensated reference signal $\hat{r}(n)$ and the error signal $e(n)$ are fed to M contiguous single-sideband bandpass filters and then downsampled by a factor D to produce M subband signals. The M contiguous single-sideband bandpass filters are realized by employing the polyphase FFT technique below [11],

$$\begin{aligned} r_m(k) &= \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} r(kD-i) \\ e_m(k) &= \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} e(kD-i) \\ \hat{r}_m(k) &= \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} \hat{r}(kD-i) \end{aligned} \tag{9}$$

where m is the subband index, a_i are the coefficients of a K_L point prototype filter, and k is the subband time index. To be concise, the subband delay compensated reference signal and the subband orthogonal delay compensated reference signal are called the subband reference signal and subband orthogonal reference signal respectively hereafter in the paper.

(4) Adaptive Weight Update: To derive the full-band adaptive filter update equation, the cost function is defined as the sum of the squared subbands error $e_m(k)$ from 0 to $M-1$,

$$J(k) = \sum_{m=0}^{M-1} A_m |e_m(k)|^2 \tag{10}$$

where A_m is the weighting coefficient for the mth subband. This definition brings flexibility in the active control algorithm for noise with different spectrum components for different control goals.

By using the steepest descent method [1], the update equation for the lth controller coefficient can be written as,

$$w_l(k+1) = w_l(k) - \mu \nabla_{w_l} \tag{11}$$

where μ is the step size, ∇_{w_l} denotes the gradient at $w = w_l$, which can be expressed as,

$$\nabla_{w_l} = \frac{\partial J(k)}{\partial w_l} = \sum_{m=0}^{M-1} A_m \left[e_m(k) \frac{\partial e_m^*(k)}{\partial w_l} + e_m^*(k) \frac{\partial e_m(k)}{\partial w_l} \right] \tag{12}$$

According to Eq. (4), $e(kD-i)$ can be represented as,

$$e(kD-i) = d(kD-i) + v(kD-i) + \sum_{l=0}^{N-1} w_l x_{\mathrm{f}}(kD-i-l) \tag{13}$$

where $x_{\mathrm{f}}(k)$ is the filtered reference signal. Substituting Eq. (13) into Eq. (9), the mth subband error signal $e_m(k)$ can be written as,

$$\begin{aligned} e_m(k) = \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} \Bigg[& d(kD-i) + v(kD-i) \\ & + \sum_{l=0}^{N-1} w_l x_{\mathrm{f}}(kD-i-l) \Bigg] \end{aligned} \tag{14}$$

so the gradient ∇_{w_l} can be written as,

$$\begin{aligned} \nabla_{w_l} &= \sum_{m=0}^{M-1} A_m \Bigg\{ e_m(k) \sum_{i=0}^{K_L-1} a_i e^{-\mathrm{j}2\pi\frac{mi}{M}} x_{\mathrm{f}}^*(kD-i-l) \\ &\qquad + e_m^*(k) \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} x_{\mathrm{f}}(kD-i-l) \Bigg\} \\ &= 2 \sum_{m=0}^{M-1} A_m \mathrm{Re}\{e_m(k) x_{\mathrm{f},m}^*(k-l)\} \end{aligned} \tag{15}$$

where the superscript * denotes complex conjugate and the mth subband filtered reference signal $x_{\mathrm{f},m}(k)$ is obtained by,

$$x_{\mathrm{f},m}(k) = \sum_{i=0}^{K_L-1} a_i e^{\mathrm{j}2\pi\frac{mi}{M}} x_{\mathrm{f}}(kD-i) \tag{16}$$

Substituting Eq. (15) into Eq. (11), the full-band adaptive filter coefficient is updated by,

$$w_l(k+1) = w_l(k) - 2\mu \sum_{m=0}^{M-1} A_m \mathrm{Re}\{e_m(k) x_{\mathrm{f},m}^*(k-l)\} \tag{17}$$

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TABLE I
THE SIMPLIFIED SUBBAND ANC ALGORITHM WITHOUT SECONDARY PATH MODELING

Initialization stage

1. Initializing the adaptive controller coefficients (with zeros), the number of samples P used for the noise power estimation at the error sensor; the step size μ; the fluctuation factors δ, δ_1; the variation factor ν; and the step-size update factor α.

Reference signal and update direction selection stage

2. Keep the adaptive controller coefficients unchanged, measure the mean noise power with $\xi_0 = \sum_{k=0}^{P-1} e(k) * e(k)$, and the reference signal power with $\chi_0 = \sum_{k=0}^{P-1} x(k) * x(k)$.

The following steps are repeated for each subband one by one. For example, for the mth subband,

3. Without updating the adaptive controller coefficients, measure the subband mean noise power with $\xi_1 = \sum_{k=0}^{P-1} |e_m(k)|^2$, the maximum noise amplitude with $e_{\max} = \max[|e_m(k)|]$, the subband reference signal power with $\chi_1 = \sum_{k=0}^{P-1} |x_m(k)|^2$, and the subband orthogonal reference signal power with $\hat{\chi}_1 = \sum_{k=0}^{P-1} |\hat{x}_m(k)|^2$.
4. Select the delay compensated reference signal $r_m(k)$ or the orthogonal delay compensated reference signal $\hat{r}_m(k)$, and try the following step for μ or $-\mu$ respectively to find the best reference signal and update direction (4 combined candidate options).
   a. Update the adaptive controller coefficients, measure the mean noise power ξ_2, the reference signal power χ_2 and the orthogonal reference signal power $\hat{\chi}_2$ for another P samples, or stop updating if $|e_m(k)| > (1+\delta)e_{\max}$.
   b. If the convergence condition $|e_m(k)| < (1+\delta)e_{\max}$ is satisfied, then update the adaptive controller coefficients with equation (18).
5. If for μ and $-\mu$, the diverge condition $|e_m(k)| > (1+\delta)e_{\max}$ is met, then decrease the step size by using $\mu = \alpha\mu$, otherwise, increase the step size by using $\mu = \mu/\alpha$. After adjusting the step size, go back to step 3 to seek the best reference signal and update direction with the new step size.
6. Compare the mean noise power ξ_2 of the 4 options in step 4, select the best subband reference signal type and update direction that correspond to the minimum ξ_2.

Update stage

7. Update the adaptive controller with equation (18) with the selected reference signal and update direction of all subbands.

Monitoring stage

8. Initial $k = 1$, the mean noise power $\xi(0) = \xi_0$ and the mean reference signal power $\chi(0) = \chi_0$.
9. Calculate the mean noise power $\xi(k) = \lambda\xi(k\text{-}1) + |e(k)|^2$ and the mean reference signal power $\chi(k) = \lambda\chi(k\text{-}1) + |x(k)|^2$ iteratively, where the parameter λ is a forgetting factor, and is usually chosen a value between 0.9 and 0.99.
10. If $\nu\xi_0/\chi_0 < \xi(k)/\chi(k)$ or $\xi(k)/\chi(k) > (1+\delta_1)\nu\xi(k\text{-}P)/\chi(k\text{-}P)$, then go to the reference signal and direction selection stage; otherwise keep updating.

(5) Subband Filtered Reference Signal and Update Direction Selection: The key step of the proposed algorithm is the subband filtered reference signal and the update direction selection. Without the secondary path information, the filtered reference signal $x_{\mathrm{f}}(n)$ is substituted by the delayed modified reference signal $r(n)$ or the orthogonal delay compensated reference signal $\hat{r}(n)$, and equation (17) is re-expressed as,

$$w_l(k+1) = w_l(k) - 2\sum_{m=0}^{M-1} \mu_m \mathrm{Re}\{e_m(k) R_m^*(k-l)\} \quad (18)$$

where $R_m^*(k)$ is the conjugate of $R_m(k)$. $R_m(k)$ denotes $r_m(k)$ or $\hat{r}_m(k)$, $\mu_m = \mu c_m A_m$ is the step size for the mth subband with μ being the step size and c_m being the sign of the step size for the mth subband update direction.

In the proposed algorithm, the update direction can be set at either 0 or 180 degree, and the subband reference signal can be either the delay compensated reference signal or the orthogonal delay compensated reference signal, where the delay compensated reference signal is abbreviated as DCR while the orthogonal delay compensated reference signal is abbreviated as ODCR. It has 4 update options for each subband: (i) the DCR and 0° update direction; (ii) the DCR and 180° update direction; (iii) the ODCR and 0° update direction; and (iv) the ODCR and 180° update direction. If the phase response difference between the update direction and the secondary path at the mth subband is larger than 90°, the DCR $r_m(k)$ and the update direction 180° are selected. However, if the phase response difference between the update direction and the secondary path at the mth subband is close to $\pm 90°$, the ODCR $\hat{r}_m(k)$ is selected. All selections are based on monitoring the residual mean noise power in each subband. The initial sign of step size can be assumed to be -1, and the initial subband filtered reference signal can be assumed as $r_m(k)$.

The procedure of the whole algorithm consists of 4 stages, which are detailed in Table I, where the parameter P is the number of samples used for the noise power estimation at the error sensor, and the selection of its value depends on the dynamic characteristics of the primary noise. Large P increases the reliability of the estimation but requires longer estimation time. The fluctuation factors δ, δ_1 and the variation factor ν are introduced to make the algorithm tolerate the estimation error of residual noise power while remaining sensitive to changes in the secondary path, which are initialized in the same way as that in Wu's algorithm [12]. For stationary additive background noise, the variation factor ν can be set as $1 + \delta_1$, and δ_1 can be chosen between 0.1 and 0.5 while δ can be chosen between 0 to 0.5. The step size μ affects the stability of the proposed algorithm, and it should be in the range $0 < \mu < 2/P_x(N + \Delta)$, where P_x is the mean of the reference signal, N is the length of the controller filter, and Δ is the delay of the secondary path [1]. If a large μ is chosen, the proposed algorithm may diverge; however, it takes more time to choose the proper reference signal and search the best update direction for all subbands when a small step size μ is chosen. The step-size update factor α is a factor for adjusting the step size, which is usually chosen to be 0.5.

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Fig. 3. A duct with different acoustic end conditions (a) absorption end; (b) open end; (c) rigid end.

III. Simulations and Experiments

The primary path $P(z)$ and the secondary path $S(z)$ used in the simulations are measured from a duct with 3 different acoustic end conditions as shown in Fig. 3, and each of them is modeled by a 256-tap FIR filter. The length of the adaptive controller is also set as 256.

The secondary path transfer function between the output of adaptive controller and the input of error signal includes the D/A converter, the power amplifier, the loudspeaker, acoustic path from the loudspeaker to the error sensor, the sensor, the preamplifier and the A/D converter [2]. The total delay of the secondary path can be expressed as,

$$\tau = \tau_1 + \tau_2 \tag{19}$$

where $\tau_1 = N_s/f_s$ is the electrical delay of the path and f_s is the sampling frequency. $N_s = 15$ is the sampling numbers when the reference signal propagates from the input to the output of the control hardware, and it can be measured by loopback of input and output of the controller. The hardware used to test the proposed algorithm is an ADSP21161 EZ-KIT Lite board and the system worked at a sampling rate of 2400 Hz. $\tau_2 = l/c$ is the acoustic delay from the secondary source to the error sensor, where l is the distance from the error sensor to the secondary source and $c = 340$ m/s is the speed of the sound in air.

In Wu's algorithm, the constraint for stable narrowband noise control is derived, and the phase range of the secondary path response in a subband is required to be less than 90° to satisfy the geometric analysis technique in Zhou's algorithm [9], i.e.,

$$\varphi_{\max} - \varphi_{\min} < 90^\circ \tag{20}$$

where $\varphi_{\max}$ is the maximum phase and $\varphi_{\min}$ is the minimum phase in a subband [10]. If this condition is satisfied, the adaptive filter coefficients are bound to move towards the global minimum. For a pure delay system, the relation between the direct path delay τ and the phase response φ is,

$$\varphi = -360^\circ \times f \times \tau \tag{21}$$

where f is the frequency, and the minus indicates the lag of the phase. In the duct system shown in Fig. 3, the distance between the primary source and the error sensor is 136 cm, and the distance l between the secondary source and error sensor is 40 cm, so the acoustic delay τ_2 is 1.2 ms. The electric delay τ_1 is 6.25 ms by using the measured N_s, thus the total delay of the secondary path is $\tau = 7.45$ ms with equation (19), and the corresponding sampling number is about 17.

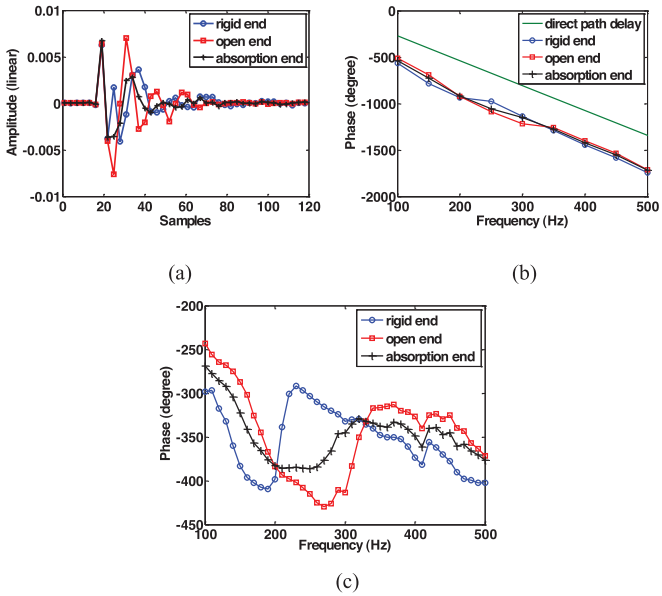


Fig. 4. The secondary paths under three different acoustic end conditions (a) the impulse response; (b) the phase response; (c) the phase response of residual secondary paths after delay compensation.

The impulse response of the secondary path can be considered to be comprised of a direct sound from the secondary source to the error sensor with a delay of 17 samples and a series of the reflections [22], [23]. The direct path delay is just one part of the practical secondary path, and the estimation of the delay need not to be accurate in the proposed algorithm. The estimated time delay is used to reduce the range of phase shift of the secondary path transfer function caused by it, so that the number of subbands can be decreased to reduce the computation load of the proposed algorithm. The delay can be estimated or measured simply by using the distance estimation (divided by speed of sound) or other existing delay estimation techniques [24], [25].

The reference signal used in the experiment is a band-limited white noise from 100 to 500 Hz, and the phase caused by the delay is $\varphi_1 = -268.2^\circ$ at 100 Hz and $\varphi_2 = -1341^\circ$ at 500 Hz according to equation (21), so the phase difference between the maximum phase and the minimum phase is 1071.8°. If the convergence condition described in equation (20) is applied in the algorithm, the signal needs to be divided into a minimum of 12 bands.

A. Secondary Path Delay Compensation

To reduce the number of the subbands, the direct sound delay can be used as an initial approximation of the secondary path. Fig. 4(a) shows the impulse responses of the secondary paths of the system in Fig. 3 which were modeled with an off-line modeling method. Fig. 4(b) shows the phase responses of the three secondary paths as well as that of the pure delay of the direct sound. After compensating the secondary path by this roughly estimated direct sound delay (removing the effect of the delay), the phase response of the remained secondary path is shown in Fig. 4(c), where the variation of the phase response

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TABLE II
REFERENCE SIGNAL AND UPDATE DIRECTION WITH DIFFERENT CONDITIONS FOR THE PROPOSED SSANC ALGORITHM AT DIFFERENT SUBBAND

Subband Range	Reference Signal Type and Update Direction		
	Absorption End	Open End	Rigid End
[60 180] Hz	DCR and 0°	DCR and 0°	DCR and 0°
[180 300] Hz	DCR and 0°	ODCR and 0°	DCR and 0°
[300 420] Hz	ODCR and 0°	ODCR and 0°	ODCR and 0°
[420 500] Hz	ODCR and 0°	ODCR and 0°	DCR and 180°

of the residual secondary path caused by different acoustic end conditions is clearly observed.

As shown in Fig. 4(b), the range between the maximum and the minimum phases of the transfer functions of the secondary paths are 1179.9°, 1201.8° and 1176.4° respectively under the three different acoustic end conditions. After compensating the impulse response of secondary path with a roughly estimated direct sound delay of 17 samples, the phase difference under the three acoustic end conditions are reduced to 118°, 197.7° and 120° respectively as shown in Fig. 4(c). To satisfy the constraint condition of equation (20), the subband number in the algorithm now only needs to be set as 4. This results in significant saving on the computation load.

B. Tracking Secondary Path Changes

In the simulations and experiments, the step size for the proposed SSANC algorithm and the FxLMS algorithm were set to be the largest possible value to assure that the adaptive controller converges at the fastest speed while remaining stable. A zero-mean white Gaussian noise is added with signal-to-noise ratio (SNR) of 30 dB in the simulations. For the SSANC algorithm, the primary broadband noise in the range of [100 500] Hz was divided uniformly into 4 subbands as [60 180] Hz, [180 300] Hz, [300 420] Hz and [420 500] Hz to account for various acoustic end conditions, and the parameters in Table I were selected as $P = 1000$, $\delta_1 = 0.2$, $\lambda = 0.9$, $\mu = 0.01$, $\nu = 1.2$, and $\delta = 0.1$.

In the first set of simulations and experiments, the tracking capabilities of both algorithms under different duct end conditions were compared. With the FxLMS algorithm, the off-line identified secondary path model under the absorption end was adopted. For the first 75 seconds, the duct was with the absorption end, and the ANC system did not turn on before the 15th second. For the SSANC algorithm, the reference signal and the update direction for each subband were identified as shown in the second column of Table II. It took about 18 seconds and 13 seconds respectively for the SSANC and the FxLMS algorithm to converge as shown in Fig. 5, where the mean square error curves were averaged by 300 trials in the simulations.

When the secondary path was changed to the open end condition at the 75th second, both the SSANC algorithm and the FxLMS algorithm began to diverge. However, the SSANC algorithm automatically adjusted its reference signal and update direction for each subband as shown in the third column of Table II. It took about 15 seconds for the SSANC algorithm to converge while the FxLMS algorithm diverged due to the large difference of the secondary path between the absorption end and the open end conditions. When the secondary path was changed to the rigid end condition at the 135th second, the reference signal and the update direction of the SSANC algorithm for each subband were identified as shown in the fourth column of Table II. It took about 20 seconds for the SSANC algorithm to converge while the FxLMS algorithm still diverged. From the 205th second to the 265th second, the secondary path was changed back to the open end condition. The SSANC algorithm showed good tracking performance again by automatically changing back its reference signal and the update direction for each subband as shown in the third column of Table II.

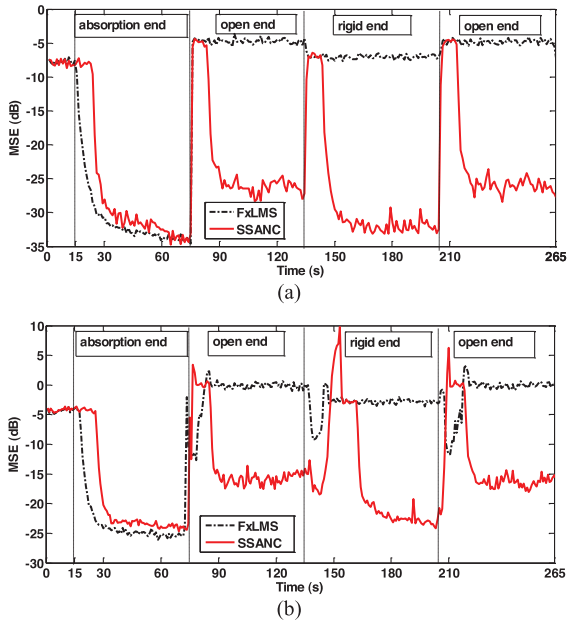


Fig. 5. The mean square error curves of the SSANC algorithm and the FxLMS algorithm with different acoustic end conditions (a) simulations; (b) experiments.

To compare the steady-state performance between the proposed SSANC algorithm and the FxLMS algorithm, the second set of simulations and experiments were conducted, in which the proper off-line identified secondary path models were utilized for the FxLMS algorithm in each state. The spectrums of the error signal with and without control after convergence are shown in Figs. 6–8, where the results are averaged by 1000 samples of the convergence error signal. The noise reduction is limited by the background noise in the experiments. For the absorption end, the noise reduction level for the FxLMS algorithm and the SSANC algorithm are 26.5 dB and 25.8 dB respectively in the simulation, and 21.1 dB and 20.6 dB respectively in the experiment. For the open end, the noise reduction level for the FxLMS algorithm and the SSANC algorithm are 22.2 dB and 19.1 dB respectively in the simulation, and 20.5 dB and 17.9 dB respectively in the experiment. For the rigid end, the noise reduction level for the FxLMS algorithm and the SSANC algorithm are 26.1 dB and 24.5 dB respectively in

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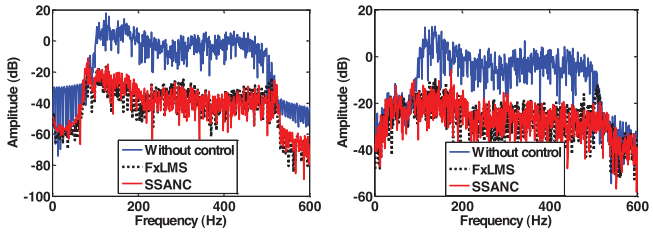


Fig. 6. Performance of the FxLMS algorithm and the SSANC algorithm under the absorption end condition (a) error signal spectrums (simulation); (b) error signal spectrums (experiment).

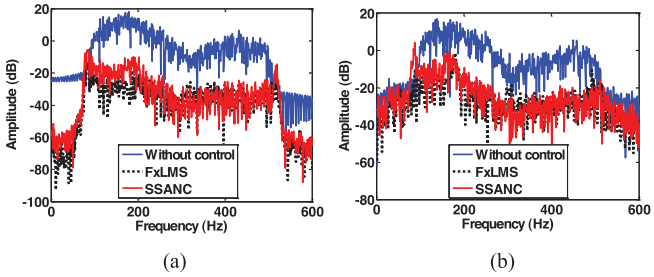


Fig. 7. Performance of the FxLMS algorithm and the SSANC algorithm under the open end condition (a) error signal spectrums (simulation); (b) error signal spectrums (experiment).

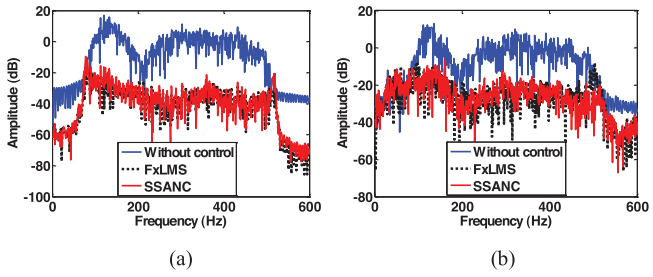


Fig. 8. Performance of the FxLMS algorithm and the SSANC algorithm under the rigid end condition (a) error signal spectrums (simulation); (b) error signal spectrums (experiment).

the simulation, and 20.8 dB and 19.5 dB respectively in the experiment. It can be seen that in most cases the SSANC algorithm has similar level of noise reduction as that of the FxLMS algorithm.

The results presented above demonstrate the effective convergence of the proposed SSANC algorithm under different secondary paths owing to the continuous monitoring of the error signal and the automatic selection of the proper reference signal and search direction in each subband. The FxLMS algorithm does not converge when there is a large change in the secondary path if there is no online secondary path modeling employed. The SSANC algorithm can achieve similar noise attenuation as the FxLMS algorithm after convergence.

The tracking performance of the SSANC algorithm was also compared to that of a typical on-line secondary path modeling method proposed by Akhtar [6]. The reference signal used in the simulation is a band-limited pink noise from 100 to 500 Hz. A zero-mean white Gaussian noise is added with signal-to-noise ratio (SNR) of 30 dB. For the SSANC algorithm, the

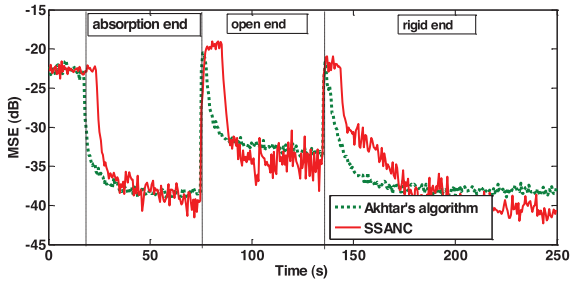


Fig. 9. The mean square error curves of the SSANC algorithm and the Akhtar's algorithm with different acoustic end conditions.

TABLE III
REFERENCE SIGNAL AND UPDATE DIRECTION WITH DIFFERENT CONDITIONS FOR THE PROPOSED SSANC ALGORITHM AT DIFFERENT DISTANCE BETWEEN THE SECONDARY SOURCE AND ERROR MICROPHONE

Subband Range	Reference Signal Type and Update Direction		
	34 cm	40 cm	66 cm
[60 180] Hz	ODCR and 0°	DCR and 0°	ODCR and 0°
[180 300] Hz	DCR and 180°	ODCR and 0°	ODCR and 0°
[300 420] Hz	DCR and 180°	ODCR and 0°	DCR and 180°
[420 500] Hz	ODCR and 180°	ODCR and 0°	DCR and 180°

primary broadband noise in the range of [100 500] Hz was again divided uniformly into 4 subbands as [60 180] Hz, [180 300] Hz, [300 420] Hz and [420 500] Hz to account for various acoustic end conditions, and the parameters in Table I were selected as $P = 1000$, $\delta = 0.3$, $\lambda = 0.95$, $\mu = 0.8$, $\nu = 1.5$, and $\delta_1 = 0.5$. The reference signal and the update direction of the SSANC algorithm for each subband were also identified as shown in the Table II. For the Akhtar's algorithm, a zero-mean white Gaussian noise of variance 4×10^{-4} was used in the modeling process. The learning curves of the two algorithms are shown in Fig. 9. It can be seen that both algorithms can track the secondary path variation. The convergence speed of the SSANC algorithm is slower than that of the Akhtar's algorithm due to the update direction searching process. However, the steady-state residual noise is lower than that of the Akhtar's algorithm since no auxiliary noise is needed in the SSANC algorithm.

The tracking performance of the proposed SSANC algorithm was also tested under different direct path delay conditions. In the simulation, the reference signal is still a band-limited pink noise from 100 to 500 Hz, and the direct path delay varies in the open end condition with different distance between the error sensor and the secondary source of 34 cm, 40 cm and 66 cm respectively. The parameters in Table I were selected as $P = 1000$, $\delta = 0.3$, $\lambda = 0.99$, $\mu = 0.7$, $\nu = 1.5$, and $\delta_1 = 0.5$. The reference signal and the update direction of the SSANC algorithm for each subband were identified as shown in Table III.

If the distance between the error sensor and the secondary source is known, the direct path delay can be estimated. Then

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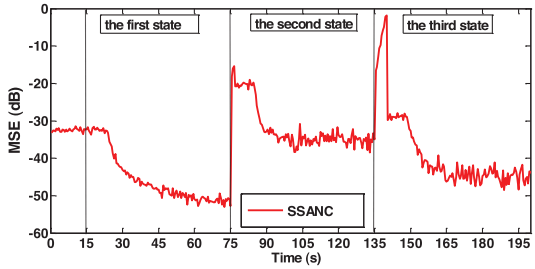


Fig. 10. The mean square error curves of the SSANC algorithm with different direct path delays.

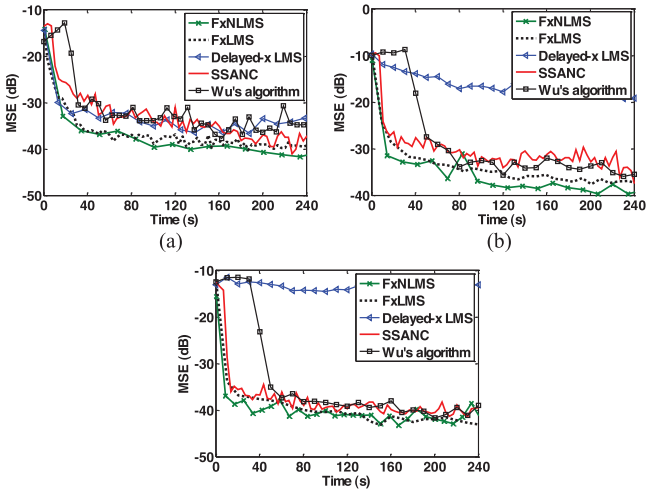


Fig. 11. The mean square error curves of the FxLMS algorithm, the FxNLMS algorithm, the delayed-x LMS algorithm, the SSANC algorithm and Wu's algorithm under different end conditions (a) absorption end; (b) open end; (c) rigid end.

the SSANC algorithm can work well under different direct path delays. The learning curve is shown in Fig. 10. It can be seen that the SSANC algorithm has good tracking performance with varied direct path delays.

C. Comparing with Other ANC Algorithms

In this section, the performance of the proposed SSANC algorithm, the FxLMS algorithm, the normalized FxLMS (FxNLMS) algorithm, the delayed-x LMS algorithm and Wu's algorithm are compared via simulations. In the simulations, the secondary path models were modeled correctly off-line first and then were adopted for the FxLMS algorithm and the FxNLMS algorithm at each state. Fig. 11 shows the learning curves of the 5 algorithms under different end conditions. It can be clearly observed that the FxNLMS algorithm has the fastest convergence rate. Wu's algorithm does not compensate for the secondary path delay, so significantly more subbands are required to satisfy Eq. (20) in each subband. The subbands number are 11, 15, and 18 under the three different end conditions respectively for Wu's algorithm, while only 4 subbands are needed for the SSANC algorithm. Note that the appropriate updating direction has to be determined in each subband in turn,

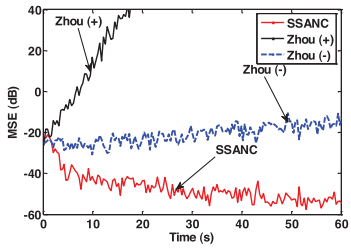


Fig. 12. The mean square error curve of Zhou's algorithm (+ sign), Zhou's algorithm (− sign) and the SSANC algorithm.

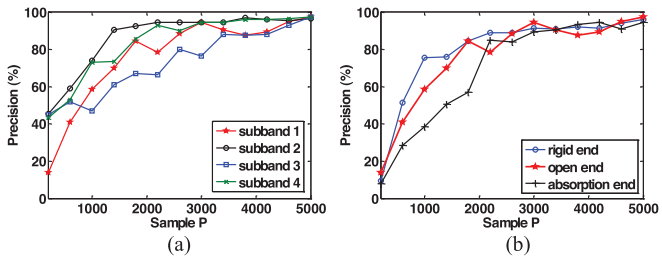


Fig. 13. Estimation precision as a function of P with (a) four subbands under the open end condition; (b) the first subband under the three different acoustic end conditions.

therefore more subbands will lead to more convergence time. Consequently, the SSANC algorithm converges significantly faster than Wu's algorithm.

For the delayed-x LMS algorithm, the direct path delay is set $\Delta = 17$. As shown in Fig. 11, the delayed-x LMS algorithm has good performance under absorption end conditions; however, it does not work well under the open end and the rigid end conditions. This is because the direct path delay is used as the estimated secondary path model in the delayed-x LMS algorithm and the mismatch between the estimated secondary path model and the true secondary path is large when the end condition changes to rigid or open. For the proposed SSANC algorithm, the proper reference signal is selected and the best update direction is searched in each subband so it can approximate the phase of the residual secondary path and works well under all end conditions.

Fig. 12 shows the performance of Zhou's algorithm and the SSANC algorithm for the narrowband primary noise in the range of [220 228] Hz. For Zhou's algorithm, whatever the sign in front of the step size, the algorithm dose not converge for a long time because the phase difference of the frequency range is close to 90° under this specific situation. However, the SSANC algorithm converges well with the orthogonal reference signal and the 0° update direction.

D. Selection of Parameters P and Step Size μ

The parameter P is the number of samples used for the noise power estimation at the error sensor, and the selection of its value depends on the dynamic characteristics of the primary noise. Large P increases the reliability of the estimation of the reference signal type and the update direction, but requires longer estimation time. The optimal value of P is chosen by

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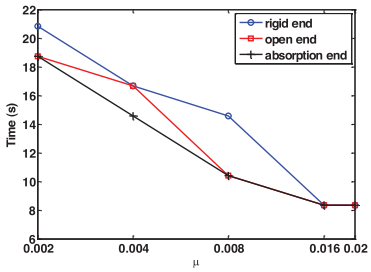


Fig. 14. The total time for choosing the proper reference signal and searching the best update direction with different step size μ under the three different acoustic end conditions.

trial and error in this paper. Fig. 13(a) shows the estimation precision as a function of P for different subbands under the open end condition while Fig. 13(b) shows the precision of the estimation as a function of P of the first subband under three different acoustic end conditions. All the precision were calculated based on 200 trails in the simulation. When the parameter P is changed from 600 to 800, the precision increases from 24.5% to 41% for the first subband, and the precision increases from 47.5% to 59% for the second subband. The precision for the other two subbands is more than 50%.

In order to reduce the estimation error, the number P needs to be larger than 600. For $P = 1000$, the precision is 58.5%, 74%, 73% for the first subband, the second subband and the fourth subband respectively. The precision for the third subband is 47%, close to 50%. For $P > 1000$, the precision is increased for all the 4 subbands, however, it takes much more time to choose the proper reference signal type and search the best update direction for the 4 subbands because more samples are needed to estimate the noise power. For $P < 1000$, the precision is reduced for the 4 subbands, which also leads to more search time for the proper reference signal and the update direction. As shown in Fig. 5, $P = 1000$ was selected in the above simulations and experiments.

To ensure the stability of the proposed algorithm, the step size μ should be in the range $0 < \mu < 2/P_x(N + \Delta)$, where P_x is the mean of the reference signal, N is the length of the controller filter, and Δ is the delay of the secondary path [1]. The length of the adaptive filter $N = 256$, the direct path delay $\Delta = 17$, $P_x = 0.0435$ for the first narrowband under the three different end acoustic condition, so the upper bound of step size $\mu = 0.1684$. To ensure the stability of the system, it starts with a value of μ on the order of a tenth of the upper bound. With the proposed SSANC algorithm, the largest step size under current situations was set at $\mu = 0.02$. The effect of the step size μ is verified under the three different end acoustic conditions, and the proper reference signal and the best update directions for the four subbands are shown in the Table II. In the simulation, the parameters $P = 1000$, $\delta = 0.2$, and $\lambda = 0.95$. For the initial step size $\mu = 0.002$, $\alpha = 0.5$, the total time for the 4 subbands to choose the proper reference signal and search the best update direction is shown in Fig. 14. It can be observed that it takes more time to choose the proper reference signal and search the best update direction for all the subbands when a small step size μ is chosen.

TABLE IV
THE COMPUTATIONAL COMPLEXITY

Operation	SSANC	Wu's algorithm
Control signal generation	N	N
Delay reference signal generation	Δ	None
Orthogonal reference signal generation	L	None
Subband signal generation	$3(K_L+M\log_2 M)/D$	$2(K_L+M\log_2 M)/D$
Frequency subband signals	None	$(4N/D)\log_2(2N/D)$
Subband control filter coefficients	None	$8N/M+16N/M^2$
Subband/fullband transform	None	$2\log_2 N+2$
Reference signal and update direction selection stage.	$M_a(10+8N)$	$(M/2+1)(32N/D+10)$
Control filter update	$M_a(2N+1)/D$	None
Total	$N+L+\Delta+M_a(2N+1)/D+3(K_L+M\log_2 M)/D+M_a(10+8N)$	$2(K_L+M\log_2 M)/D+(4N/D)\log_2(2N/D)+(8N/M+16N/M^2)+2\log_2 N+(M/2+1)(32N/D+10)+2+N$

IV. COMPUTATIONAL COMPLEXITY

In this section, the computational complexity of Wu's algorithm [12] and the proposed SSANC algorithm are compared. When the secondary path is stationary, both the two algorithms can update the adaptive filter after determining the proper reference signal and the right update direction. The computational complexity of the SSANC algorithm can be divided into six parts: (1) control signal generation; (2) DCR signal generation; (3) ODCR generation; (4) subband signal generation; (5) adaptive weight update; and (6) reference signal and update direction selection.

To avoid subband interference, the update directions for each subband is judged independently [10]. For Wu's algorithm and the proposed SSANC algorithm, the following assumptions are used, (1) the optimal step size is chosen in the process of the reference signal and update direction selection stage, so the step 5 in the Table I is not needed; (2) the convergence condition is satisfied with trials of the 4 combined candidate options. The maximum computational complexity and memory requirement for the two algorithms are shown in Table IV and Table V respectively. The length of the prototype filter is K_L, the down sampling rate is D, the number of total subbands is M, the number of the actually used subbands is M_a, the length of the control filter is N, Δ is the direct path delay, and the length of the Hilbert filter is L.

For the SSANC algorithm, $K_L = 128$, $M = 20$, $M_a = 4$, $D = 10$, $\Delta = 17$, $L = 34$, $N = 256$, then 8808 real multiplications and 2022 words are required. If the optimal reference signals and update directions are identified, only 576 real multiplications are required. For Wu's algorithm, different set of parameters are required under different acoustic end conditions: (1) for the absorption end condition, $K_L = 256$, $M = 64$, $M_a = 11$, $D = 32$, $N = 256$, $P = 1024$, then 9253 real multiplications and 3344 words are required. (2) For the open end condition, $K_L = 280$, $M = 80$, $M_a = 15$, $D = 40$, $N = 280$, $P = 1024$, then 10057 real multiplications and 3132 words are

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TABLE V
THE MEMORY REQUIREMENT

Operation	SSANC	Wu's algorithm
Coefficients of the control filter	N	N
Coefficients of the control filter in frequency-domian	None	$2N$
Coefficients of delay filter	Δ	None
Coefficients of the Hilbert filter	L	None
Prototype filter	None	K_L
Coefficients of subband filters	$2MK_L$	None
Subband control filter coefficients in frequency-domain	None	$2*(D+1)*N/D$
Subband signal generation	$3K_L$	$3MN/D$
Reference signal	N	N
Delay reference signal	Δ	None
Orthogonal reference signal generation	L	None
Total	$2(L+\Delta+N)+(2M+3)K_L$	$4N+K_L+3MN/D$ $+2*(D+1)*N/D$

required. (3) For the rigid end condition, $K_L = 300$, $M = 100$, $M_a = 18$, $D = 50$, $N = 300$, $P = 1024$, 10770 real multiplications and 3912 words are required. Apparently, the proposed SSANC algorithm has a significantly lower computational complexity in most cases. However, it should be pointed out that if the proposed SSANC algorithm and Wu's algorithm are implemented with the same number of subbands, and the downsampling rate D of the proposed SSANC algorithm is less than that of Wu's algorithm, the complexity of Wu's algorithm could be lower than that of the proposed algorithm.

For the FxLMS algorithm, when the length of the secondary path is L, and the length of the control filter is N, $(2N + L + 1)$ real multiplications per iteration are required and $(3N + 2L)$ words are required. For $L = 256$, $N = 256$, 769 real multiplications per iteration are required and 1024 words are required. It can be seen that the maximum computational burden of the proposed algorithm is larger than that of the FxLMS algorithm due to the update direction searching process. It also should be pointed out that the maximum computational burden of the proposed SSANC algorithm is more complicated than the typical online secondary-path modeling algorithms. The main benefit of our algorithm is that more noise reduction level can be achieved since no auxiliary noise is needed in the controlling process.

V. CONCLUSIONS

A simplified subband active noise control algorithm without secondary path modeling is proposed in this paper, which does not use any secondary path model directly to determine the update direction during active control, but to search the correct reference signal and the update direction in each subband by monitoring the changes of error signal. The update direction has two choices of 0° or 180°, and the reference signal in each subband has also two choices of the delay compensated reference signal or the orthogonal delay compensated reference signal. If the secondary path phase response is near $\pm 90^\circ$, the orthogonal delay compensated reference signal is selected to avoid slow convergence; otherwise, the delay compensated reference signal is selected. The main delay of the secondary path is used in the proposed algorithm to reduce the number of subbands. Unlike the conventional subband algorithms, the full-band adaptive controller in the proposed algorithm is updated directly by narrow band signals in time domain, and this removes the necessity of using the weight transform methods. The convergence speed of the proposed SSANC algorithm is slower than the FxLMS algorithm because it does not have an explicit secondary path model but uses a process of selecting reference signal and searching direction in each narrowband. Nevertheless the proposed algorithm does tract the changes of the secondary path and maintains the stability of the system. After the right reference signal and update direction are selected, the proposed algorithm can achieve similar level of noise reduction as that of the FxLMS algorithm with the correct secondary path modeling. Furthermore, the proposed algorithm has the merits of lower computational complexity and faster convergence speed compared with the existing four-direction search method.

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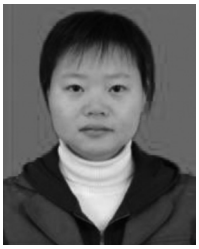
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